SIMATS ENGINEERING

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Saveetha Institute of Medical And Technical Sciences

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Engineer to Excel

An AI-Based Smart Emergency Crowd Alert System for Real-Time Crowd Density Monitoring and Risk Prediction Using Computer Vision

CSA 0808 – PYTHON PROGRAMMING

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ABSTRACT

Large public gatherings often face overcrowding and various safety risks. The proposed system uses Artificial Intelligence (AI) and Computer Vision to monitor crowds in real time. CCTV cameras capture live video streams, while the AI system detects people and estimates crowd density accurately. It can identify and predict potential crowd-related risks before emergencies occur, allowing authorities to take preventive action. When a critical situation is detected, automatic alerts are sent to security personnel and emergency responders for quick action. This solution enhances public safety, reduces the risk of crowd-related incidents, and supports efficient smart city surveillance.

INTRODUCTION

- Crowd management is a major challenge in public places and large events.
- Overcrowding can lead to stampedes, panic, injuries, and loss of life.
- Traditional monitoring relies on manual surveillance, which may miss critical situations.
- AI and Computer Vision provide automated and accurate crowd analysis.
- Real-time video analytics help identify overcrowded areas instantly.
- Predictive technology enables authorities to take preventive action before emergencies.

OBJECTIVES

- Detect and count people using AI-based Computer Vision.
- Predict crowd-related risks before emergencies occur.
- Generate automatic emergency alerts for authorities.
- Improve crowd management and reduce response time.
- Enhance public safety and support smart city surveillance.
- Monitor crowd density in real time using CCTV cameras.
- Assist security personnel in effective crowd management.
- Identify overcrowded areas through continuous video analysis.

LITERATURE SURVEY

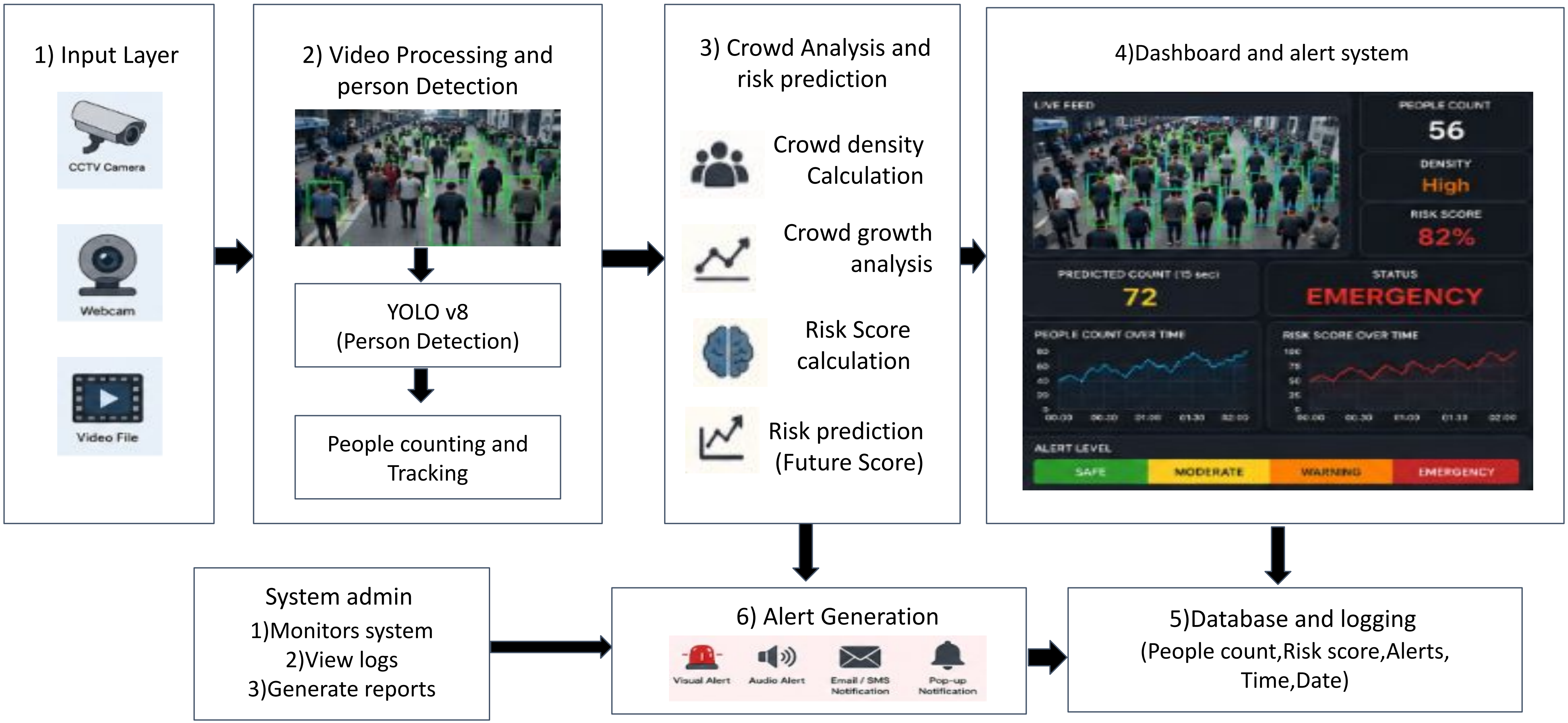
S.NO	AUTHOR NAME	PAPER TITLE	METHOD USED	DATASET	ACCURACY	FINDINGS
1	S. Balasubramanian et al.	**Proactive Headcount and Suspicious Activity Detection using YOLOv8** (2023)	YOLOv8 Object Detection	Custom Surveillance Video Dataset	95.4%	Achieved accurate real-time headcount detection and suspicious activity monitoring for public safety applications.
2	Yuchao Fang, Huanli Pang	An Improved Pedestrian Detection Model Based on YOLOv8 for Dense Scenes (2024)	Improved YOLOv8 with attention mechanism	CrowdHuman & WiderPerson datasets	96.1%	Improved pedestrian detection accuracy in crowded and occluded environments.
3	Abdullah N. Alhawsawi et al.	**Enhanced YOLOv8-Based Model with Context Enrichment Module for Crowd Counting in Complex Drone Imagery** (2024)	Enhanced YOLOv8 + Context Enrichment Module	VisDrone & DroneCrowd datasets	97.2%	Increased crowd counting accuracy in aerial images with dense crowds

PROBLEM STATEMENT

The existing traditional crowd monitoring system mainly relies on manual monitoring by security personnel, which can lead to human errors and may fail to identify overcrowded areas. Monitoring large crowds continuously is difficult, and incidents are often detected only after they occur, resulting in delayed emergency responses and alerts. The traditional system also provides limited decision support to authorities, increasing the risk of stampedes and public accidents. To overcome these limitations, the proposed **AI-Based Smart Emergency Crowd Alert System** uses Artificial Intelligence and Computer Vision for automated and continuous monitoring.

- The system analyzes live CCTV video streams, detects and counts people accurately in real time, identifies potential crowd risks before emergencies occur, and automatically generates instant emergency alerts.
- It also provides real-time insights to authorities for faster decision-making, thereby reducing crowd-related risks and improving overall public safety.

SYSTEM ARCHITECTURE



MODULE 1: CROWD DETECTION AND MONITORING

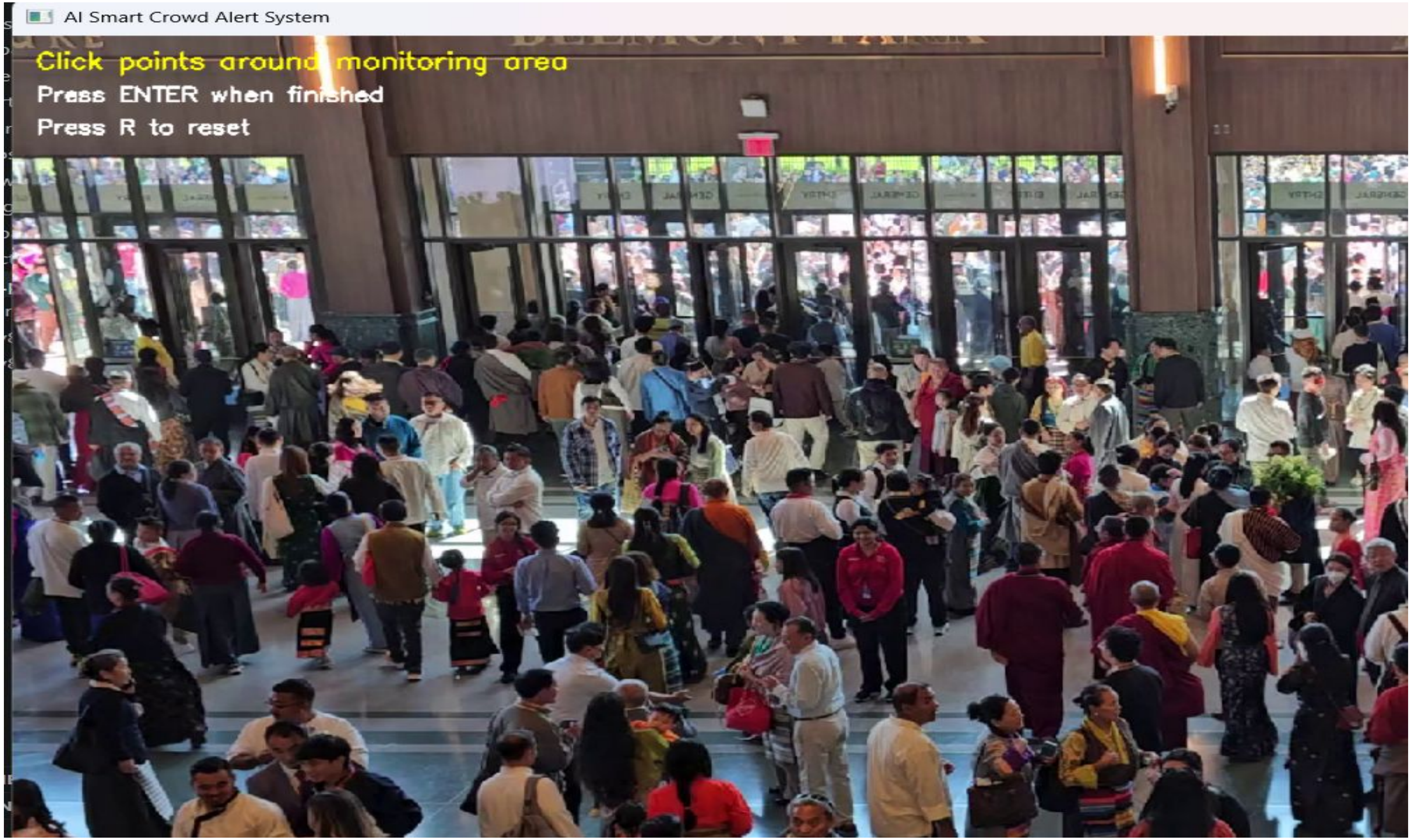
Purpose

- Detect and monitor crowd density in real time using AI and Computer Vision.
- Identify people and overcrowded areas for continuous crowd surveillance.

Functions

- Capture live video from CCTV cameras.
- Detect and count people using AI models (YOLO).
- Estimate crowd density continuously.
- Identify overcrowded areas.
- Track crowd movement for continuous monitoring.

OUTPUT


AI Smart Crowd Alert System
Click points around monitoring area
Press ENTER when finished
Press R to reset

MODULE 2: EMERGENCY RISK ANALYSIS AND ALERT SYSTEM

Purpose

- Analyze crowd conditions to identify potential emergency situations.
- Predict crowd-related risks and generate timely alerts for authorities.

Functions

- Analyze crowd density and movement patterns.
- Classify risk levels (Low, Medium, High).
- Detect abnormal crowd behavior.
- Generate automatic emergency alerts.
- Send notifications to security personnel and emergency responders.

OUTPUT

MODULE 3: DASHBOARD AND DATA MANAGEMENT

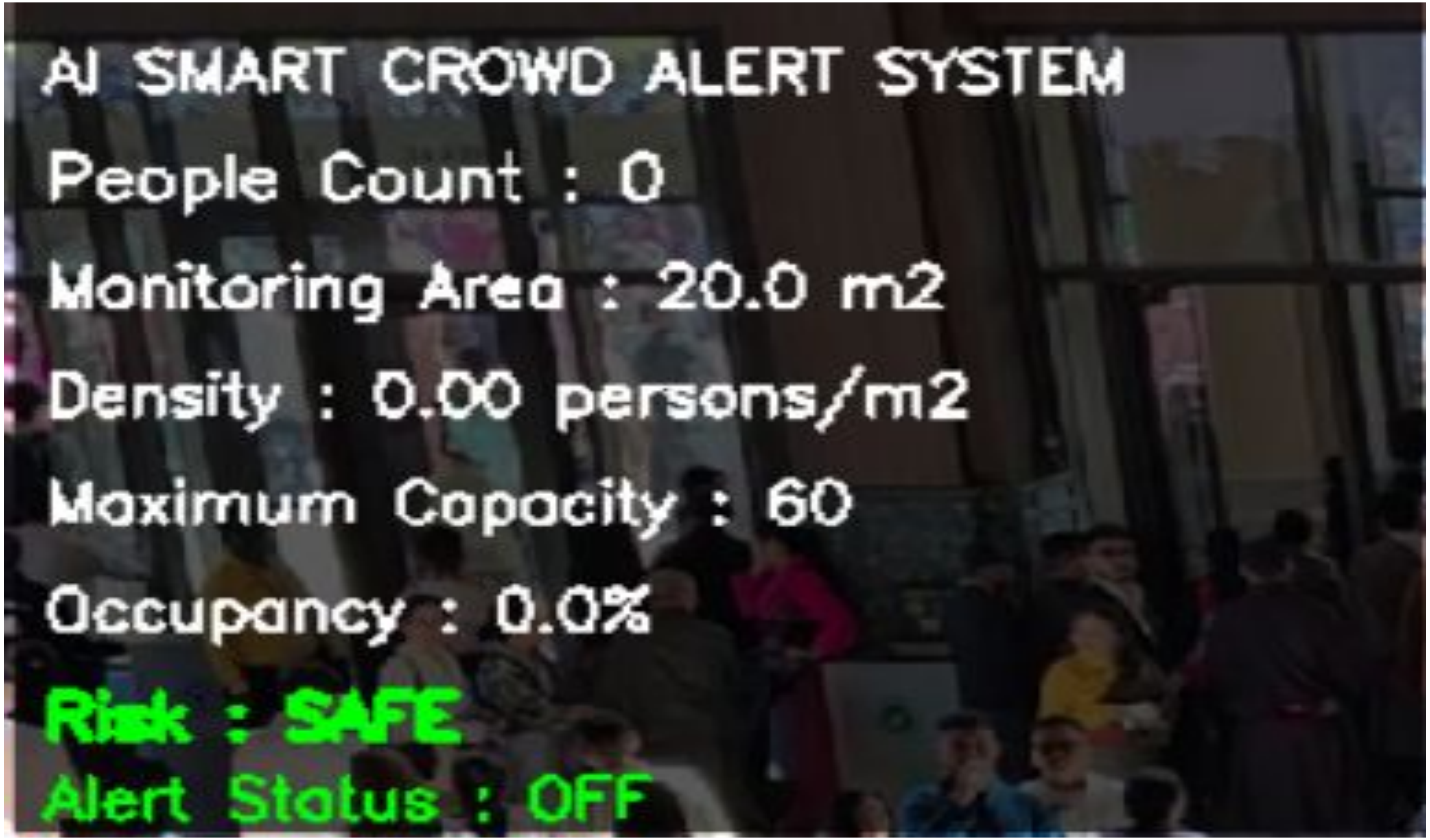
Purpose

- Provide a centralized platform for real-time crowd monitoring and system management.
- Store, visualize, and analyze crowd data to support informed decision-making.

Functions

- Display live crowd statistics and CCTV monitoring.
- Show crowd density levels and emergency risk status.
- Store historical crowd monitoring data.
- Generate reports and data analytics.
- Provide a dashboard for authorities to monitor and manage crowd situations.

OUTPUT


AI SMART CROWD ALERT SYSTEM
People Count : 0
Monitoring Area : 20.0 m2
Density : 0.00 persons/m2
Maximum Capacity : 60
Occupancy : 0.0%
Risk : SAFE
Alert Status : OFF

REFERENCES

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- Krishna Reddy, K., Ramya, T., Ramesh, Y., & Rao Pyla, J. V. G., "Real-Time Stampede Risk Prediction from Crowd Videos Using YOLOv8 and Spatio-Temporal Modeling," International Journal of Computational Science and Engineering Research, 2026.

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